

# lecture

November 10, 2025

## 1 relaxation

distortion: if a distance is 1, then it gets converted to 17 then 17 is the distortion?

### 1.1 ordinal relaxation $\alpha$

consider

$$\phi : (X, \delta) \rightarrow (Y, \delta') \quad (1)$$

$$\text{if } \alpha\delta(ij) < \delta(k, l) \implies \delta'(ij) < \delta'(kl)$$

then  $\phi$  is an ordinal embedding w relaxation  $\alpha$

i.e. distances that are different enough should be preserved (distances that are close to being the same can be violated)

## 2 tradeoffs $\alpha$ vs $d$

for every int  $d, n$ , families of metric spaces cannot be embedded unless high relaxation  $\alpha$  is allowed. i.e.  $\exists$  metric space  $T$  on  $n$  points such that the *triplet relaxation* of any ordinal embedding of  $T$  into  $d$ -dimensional euclidean space will be at least

$$\frac{\log n}{\log d + \log \log n + 8} - 1 \quad (2)$$

**Definition 1** (girth). size of the smallest cycle in a graph

the idea: create a large number of metric spaces using one graph with large girth. the number of metrics here will be greater than the number of ordinal embeddings in  $\mathbb{R}^d$ , thus cutting just one will cause the distance between some two points to be at least the girth minus one, which means the distortion/relaxation has to be at least  $G - 1$ .

key question: How dense can a graph be with a high girth?

*Proof.* pick a high girth graph  $G(V, E)$  with  $|V| = n$ .

$$m = |E| \geq \frac{1}{4} n^{1+\frac{1}{g}} \quad (3)$$

idea, start w random graph, delete edges that create small cycles. the number of edges needed to be removed is “not htat many”

pick  $g = \frac{\log n}{\log d + \log \log m + 8}$

the number of  $|E| > 16nd \log n \approx n \log n$

## 2.1 idea

now subsample the graph. large number  $N$ ,  $G_1, \dots, G_n$  ( $G_i$  are graphs)

for a pair  $G_i, G_j$ ,  $\exists v \in V$  ST in  $G_i$   $vw$  are connected, but in  $G_j$  they are not connected. but also another pair  $v, u$  are connected in  $G_j$  but not in  $G_i$ .

i.e.  $E_{G_j}(v) \setminus E_{G_i}(v) \neq \emptyset$  and  $E_{G_i}(v) \setminus E_{G_j}(v) \neq \emptyset$  where  $E(v)$  are the edges around  $v$ .

$v$  is called the “witness”. in a sense,  $v$  is a vertex where  $G_{ij}$  differ.

## 2.2 continue

now lets do the subsampling property.

independently sample each edge in the original graph w probability  $\frac{1}{2}$  w  $m$  edges. this generates  $N = 2^{bm}$  graphs for  $b < \frac{1}{2} \log_2 \frac{4}{3}$ .

**Lemma 1.** the set of subgraphs  $G_i$  satisfy the faraway property in the idea section.

*Proof.* simplifying assumptions:

- the original graph (the one w high girth) is  $k$  regular.
- independence on  $v$  being a witness for any 3 sets of  $G_{ijk}$

the probability of  $v$  being not a witness for  $G_i, G_j$  is  $\leq 2 \left(\frac{3}{4}\right)^k$

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